

Grammaticalization in Kinyarwanda mediated by age and gender despite ideologies about region

Adam An & Jean Paul Ngoboka

Today we're
talking about...

What **morphosyntactic variation**
does Kinyarwanda have?

Today we're talking about...

Who says what differently?

Is there **ongoing change**?

What **facilitates** change?

Kinyarwanda

Bantu; ~10 million speakers
national lg. of Rwanda (East Africa)
agglutinative morphology



(1) Ba-aá-ra-gi-sóm-e-ye.

3pl.SBJ-DIST\PST-DJ-OBJ-DIST\read-APPL-PFV.PST

'They were reading it for someone (yesterday or earlier).'

“disjoint” marker (DJ)

(1) Ba-aá-**ra**-gi-sóm-e-ye.

3pl.SBJ-DIST\PST-DJ-OBJ-DIST\read-APPL-PFV.PST

‘They were reading it for someone (yesterday or earlier).’

(Ngoboka & Zeller 2017,
van der Wal 2017) 6

“disjoint” marker (DJ)

(1) Ba-aá-**ra**-s

3pl.SBJ-DIST\PST-DJ-OBJ-

‘They were reading it for so

**distribution of disjoint
marker depends on...
a lot!**

information structure
TAM
constituency of vP
polarity
embeddedness...

(Ngoboka & Zeller 2017, van der Wal 2017,
Halpert 2012, Nshemezimana & Bostoen
2017)

r 2017,
al 2017)

ra- often encodes
near future / present progressive
(but not always)

(2) Ba-**ra**-gi-som-er-a.

3pl.SBJ-FUT.near-OBJ-read-APPL-IPFV

'They are about to read it.'

previous fieldwork:

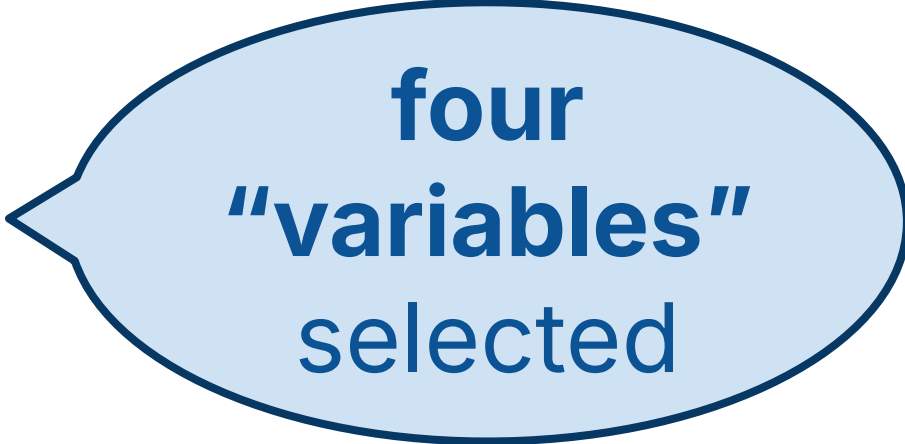
(An & Umuhoza 2023, An & Ngoboka 2025)

ra- use **varies**

previous fieldwork:

(An & Umuhoza 2023, An & Ngoboka 2025)

ra- use **varies**



four
"variables"
selected

1

near future / present progressive *ra-*
may be present or absent in:

negation

(3) **Nti**-ba-(**ra**)-som-á.

NEG-3pl.SBJ-FUT.near-NEG\read-NEG\IPFV

'They are not about to read.'

1

near future / present progressive *ra-*
may be present or absent in:

relativization

(4) *a-b-áana* *ba-(ra)-som-á*

AUG-2-children

3pl.SBJ-FUT.near-REL\read-REL\IPFV

'children who are not about to read'

1

near future / present progressive *ra-*
may be present or absent in:

participials

(an embedded category selected by a few lexical heads)

(5) N-saang-a bá-(ra)-som-a.

1sg.SBJ-CJ\realize-IPFV

PART\3p|SBJ-FUT.near-read-IPFV

'I realize they are about to read.'

2

ra- may be obligatory or optional before *ngo* 'that'

(6) N-a-(*ra*)-vúz-e *ngo* haanze híijimye.

1sg.SBJ-PST-DJ-say-PFV that it's dark outside

'I said that it's dark outside.'

3

present progressive *ra-*
competes with a **periphrastic**

(7) Ba-**ra**-som-a.

3pl.SBJ-PRES.PROG-read-IPFV

'They are reading.'

Ba-ri

3pl.SBJ-COP

gu-som-a.

INF-read-IPFV

4

ra- encodes
near future or present progressive

(8) Ba-**ra**-som-a.

3pl.SBJ-**FUT.near~PRES.PROG**-read-IPFV

'They are about to read ~ they are reading.'

1

near future / present
progressive *ra-*
may be present or absent in:
**negation, relativization,
participials**

2

ra- may be obligatory or
optional before
***ngo* 'that'**

3

present progressive *ra-*
competes with a
periphrastic

4

ra- encodes
**near future or present
progressive**

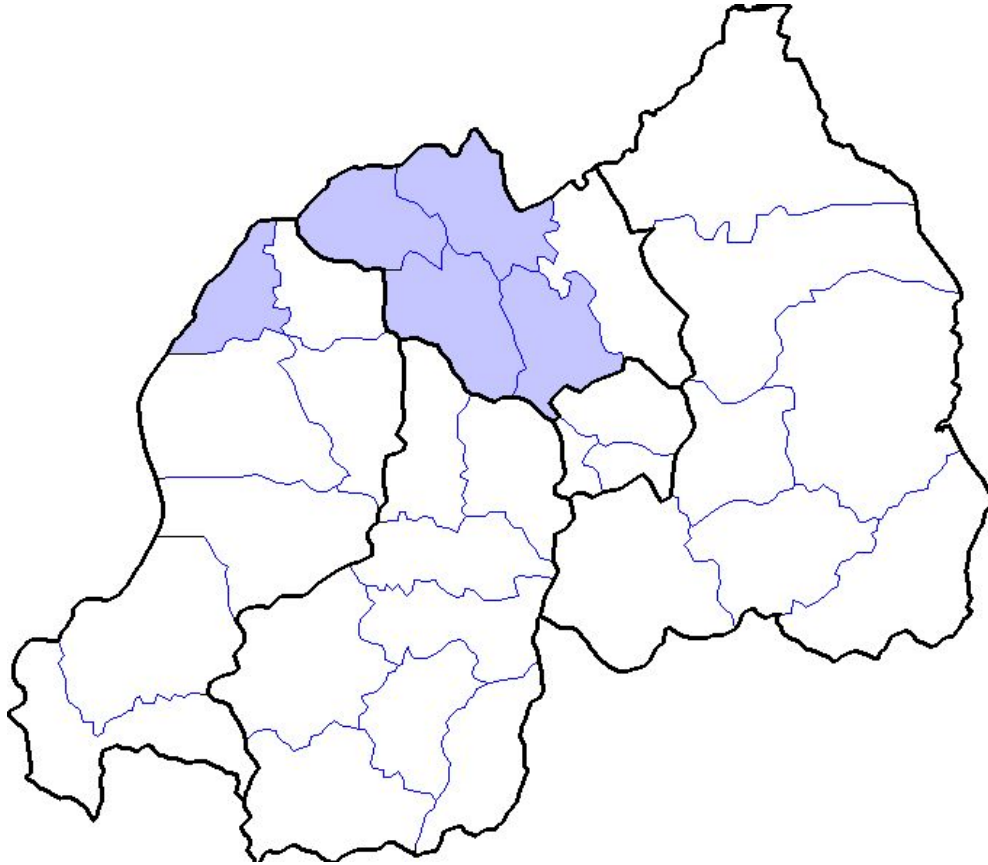
Who do
Rwandans **think**
use *ra*-
differently?

An & Ngoboka 2025:
expectations / ideologies
about these variables

(N = 15)

	aware at all	<i>ra-</i> is young	<i>ra-</i> is Northwestern	<i>ra-</i> is used by any gender
negated <i>ra-</i> 	13	4	3	0
<i>ra-</i> before <i>ngo</i> 	11	4	1	0
periphrastic 	13	3	1	0

Rwandans think 
are **Northwestern**
and **young**



"NW dialects"

ikirera, ikigoyi

stigmatized

Who actually **accepts** *ra-* differently?

survey of 65 respondents

ages 23-59

all five regions of Rwanda

17 self-reported users of NW
dialects

Who actually **accepts** *ra*- differently?

Likert scales for sentences with /
without *ra*- (White & Roberts 2022)

(worst) 1 - 5 (best)

**Morphological Preference Score
(MPS)**

score with *ra*- minus score without *ra*-

max MPS 4 = max preference **for** *ra*-

min MPS -4 = max preference **against**

Who actually
accepts *ra*-
differently?



1

near future / present
progressive *ra-*
may be present or absent in:
**negation, relativization,
participials**

1
NEG

1
REL

1
PART

crosslinguistic evidence:
ra- is **innovative** in these environments

→ **change in favor of *ra-*?**



showed preference for *ra-*
(Likert scores > 0 after scaling within participant)

64%



61%

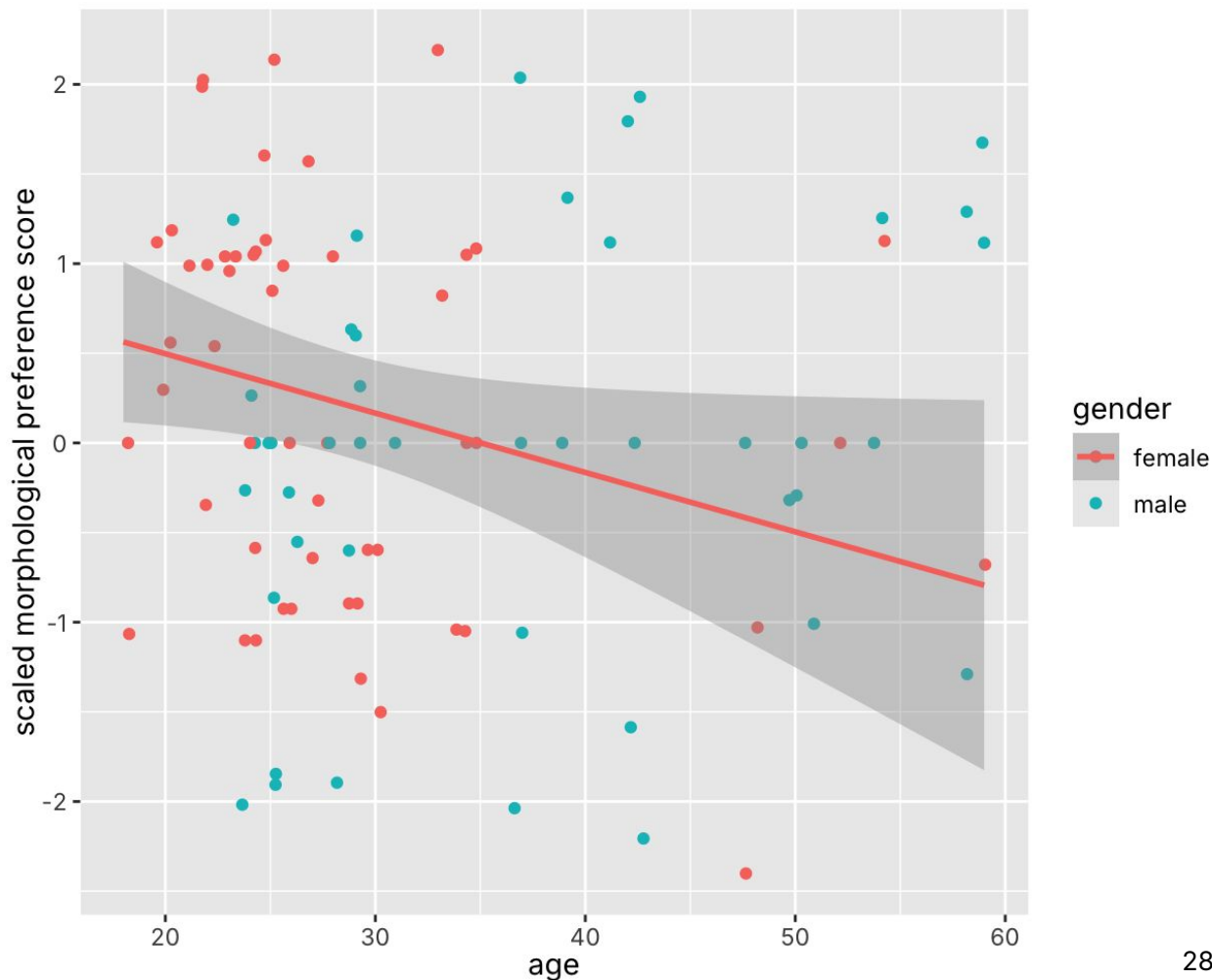


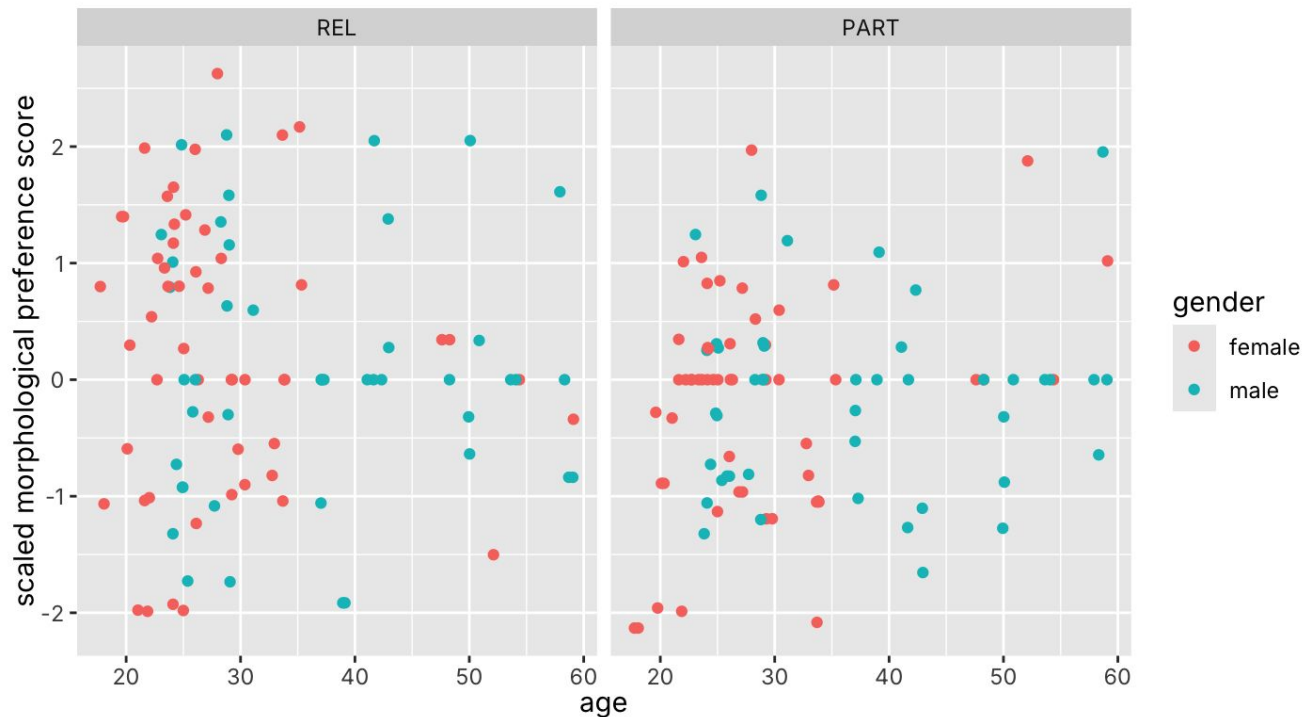
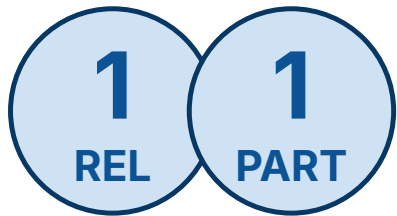
41%



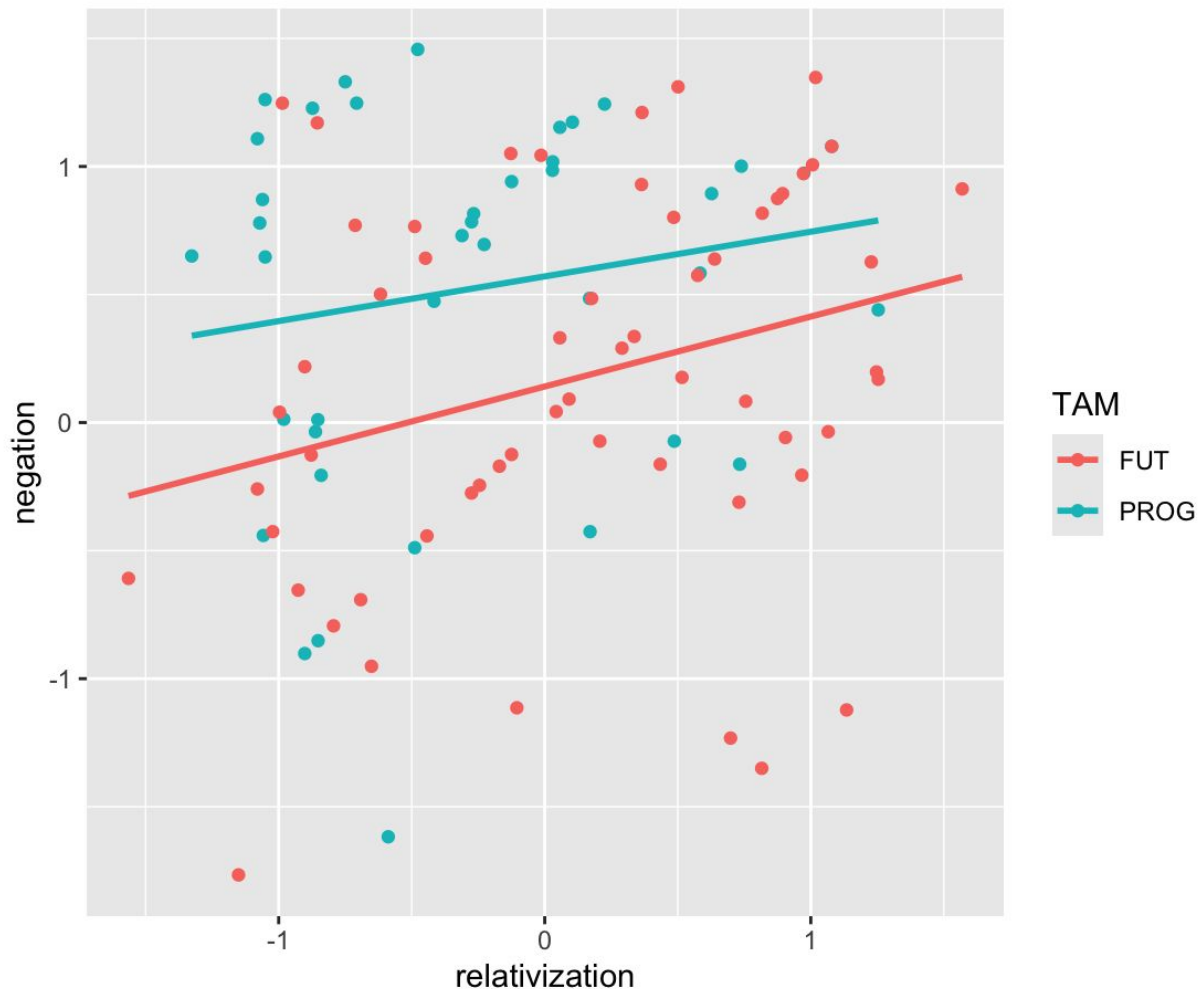
NEG

**young women
prefer *ra-* in
negation**





**no effects of
age, gender,
region, or
dialect**



post-hoc:

higher scores for
ra- in relativization
predict higher
scores in negation
($\beta = 0.273$, $p = 0.026$)

change in favor of
ra- led by young
women

stable variation?

64%



61%



41%



2

ra- is obligatory or
optional before
ngo 'that'

2

women

($\beta = -1.449$, $p = 0.060$)

young women

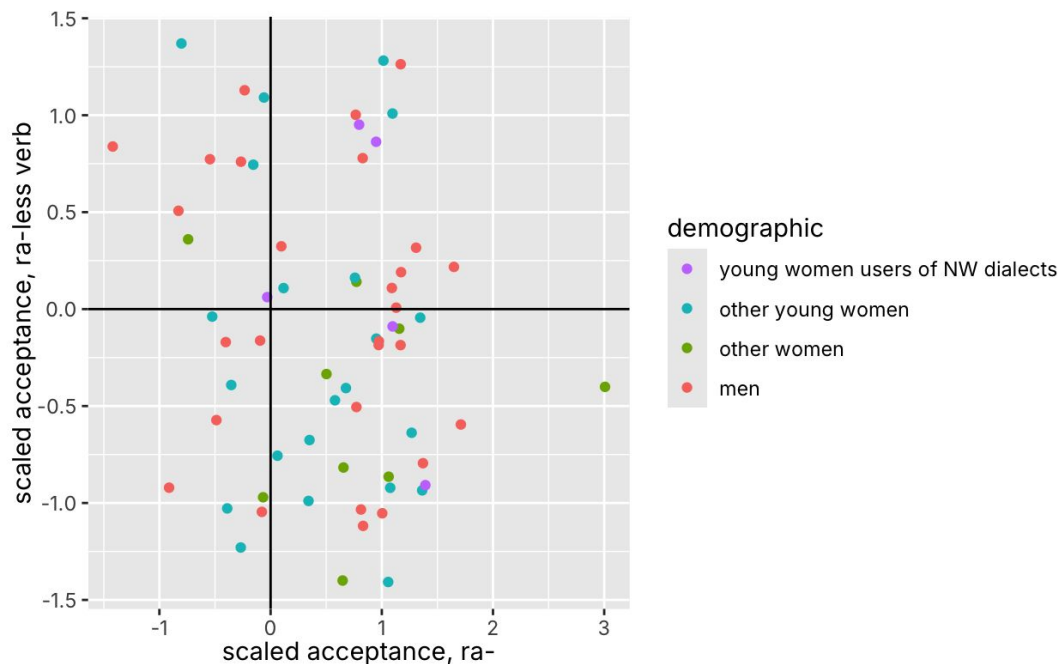
($\beta = 0.046$, $p = 0.039$)

young women users of NW
dialects

($\beta = -0.09$, $p = 0.034$)

all rate *ra*- higher

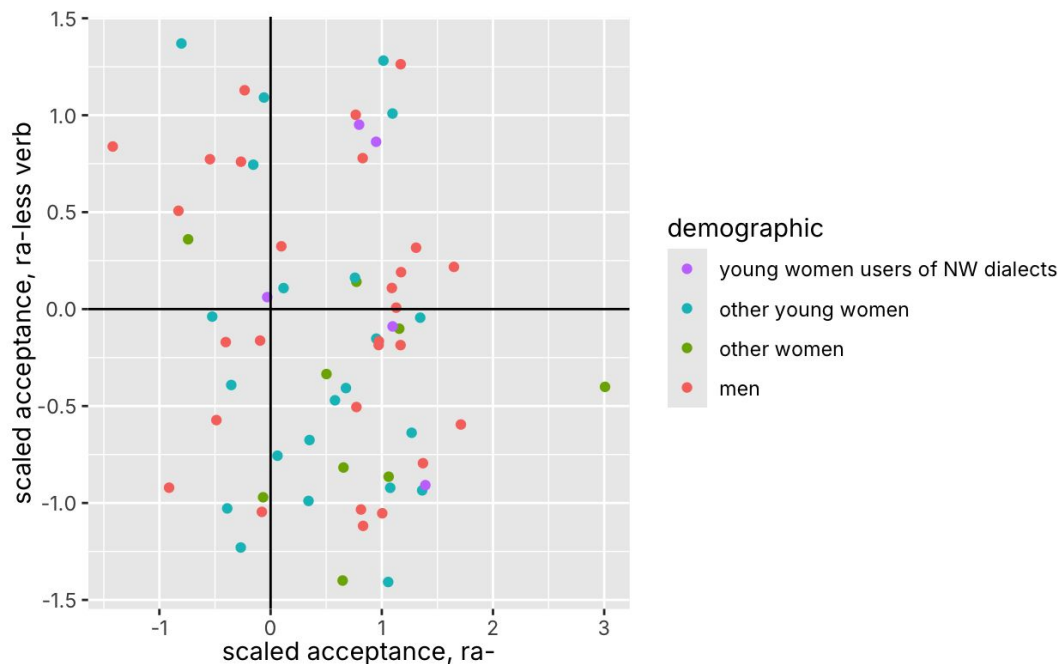
no patterns re: *ra*-less counterparts



2

**optional *ra-* becoming
more acceptable
but not more obligatory**

change led by **young
women who use NW
dialects**



1

near future / present
progressive *ra-*
may be present or absent in:
**negation, relativization,
participials**

**young women, *ra-* in
negation only**

2

ra- may be obligatory or
optional before
***ngo* 'that'**

**young women users of NW
dialects, *ra-* more preferable**

3

present progressive *ra-*
competes with a
periphrastic

4

ra- encodes
**near future or present
progressive**

1

near future / present
progressive *ra-*
may be present or absent in:
**negation, relativization,
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**young women, *ra-* in
negation only**

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ra- may be obligatory or
optional before
***ngo* 'that'**

**young women users of NW
dialects, *ra-* more preferable**

3

present progressive *ra-*
competes with a
periphrastic

no effects of age, gender, region, or dialect

4

ra- encodes
**near future or present
progressive**

Discussion

Rwandans think 
are **Northwestern**
and **young**

role of Northwest?

no clear pattern

sporadic effects, small sample size,
stigmatization → underreporting

In the IND-DP frame there was a significant interaction between gender and region ($\beta = 5.962$, $p = 0.046$) such that male Northwesterners preferred periphrastics; the interaction was attenuated by increasing age in a significant interaction ($\beta = -0.181$, $p = 0.027$) such that young male Northwesterners preferred periphrastics. In the REL frame there was a marginal main effect of gender ($\beta = 1.186$, $p = 0.098$) such that men preferred periphrastics; this effect was attenuated by region in a marginally significant interaction ($\beta = -5.514$, $p = 0.091$) such that male Northwesterners dispreferred periphrastics. In the PTCP frame there was a significant main effect of gender ($\beta = -2.185$, $p = 0.025$) such that men dispreferred periphrastics. This effect was attenuated by increasing age ($\beta = 0.058$, $p = 0.042$) such that young male Northwesterners preferred periphrastics.

evidence for change
led by young women

evidence for **stability**

all three crosslinguistically innovative, but...

change



stability



proposal one:

early change →
stability



proposal one:

later change
(picked up by our
sample)



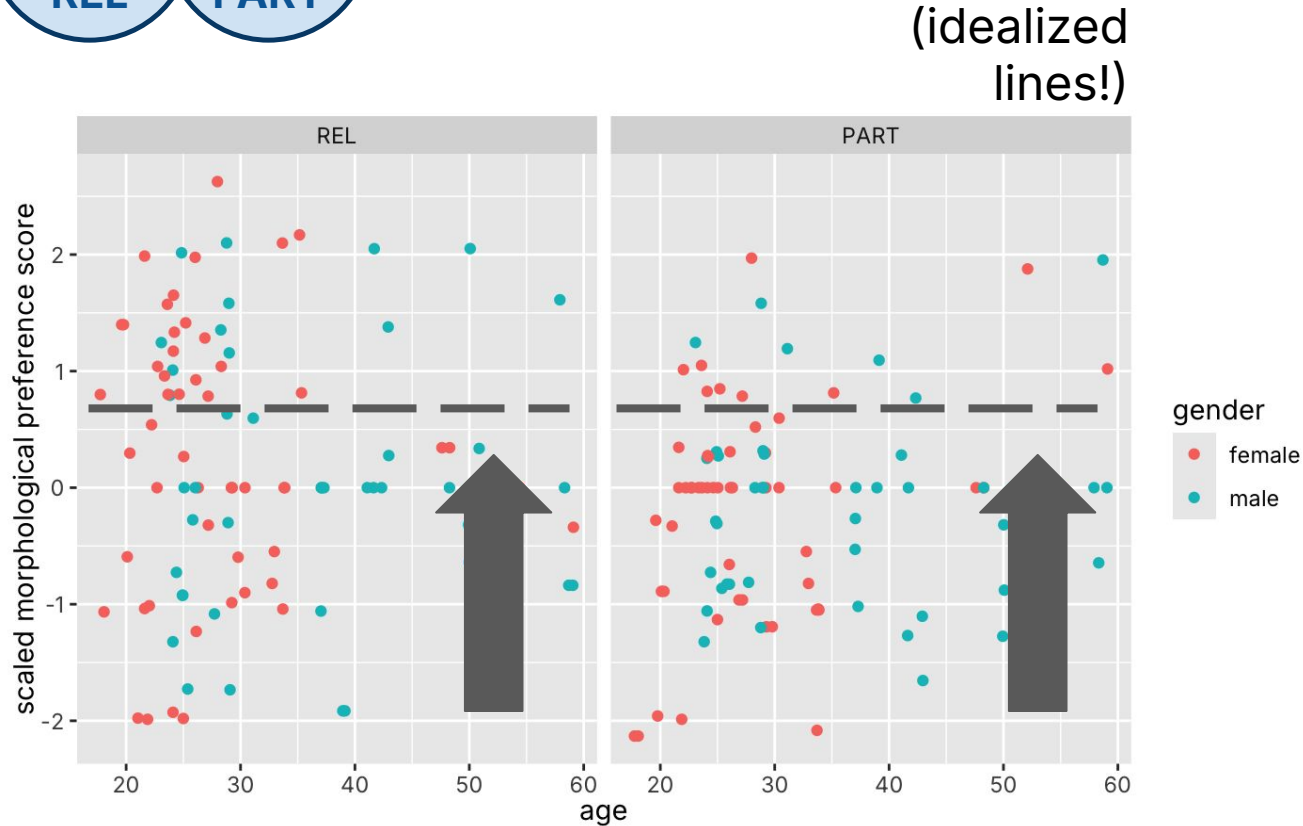
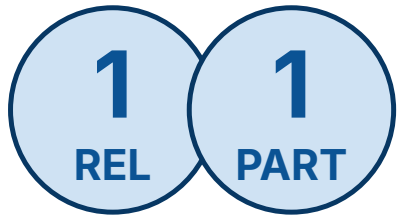
early change →
stability



proposal two:

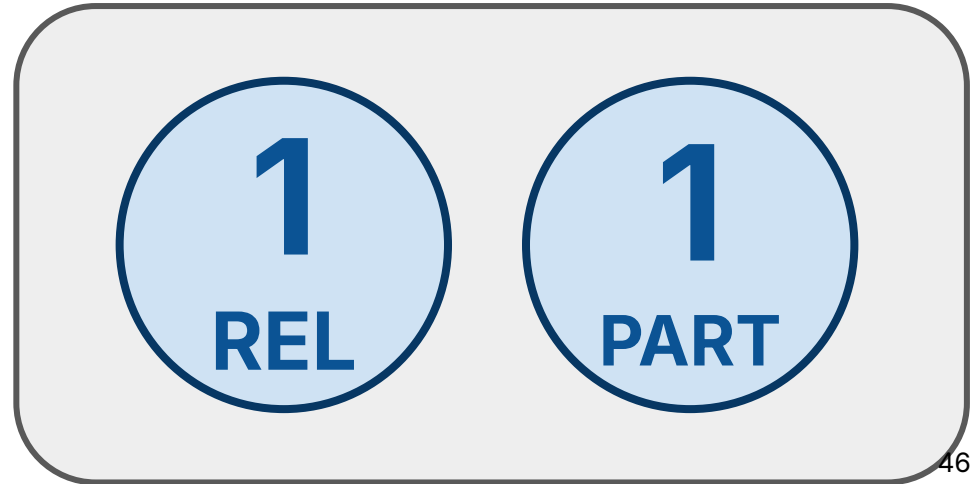
community change

adults in communities
undergoing change may
"participate in the change
themselves" (Sankoff 2019)



**raised
acceptance
among older
people
removes age
effects**

both proposals:
more susceptible
to change. why?



(9a) unmarked

*ba***ra***soma*

original distribution
(near-universal acceptance)

(9a) unmarked

*ba***ra***soma*

(9b) REL

*ba***(ra)***somá*

(9c) PART

*bá***(ra)***soma*

surface similarity
facilitates analogy

(9a) unmarked

*ba***ra***soma*

(9b) REL

*ba***(ra)***somá*

(9c) PART

*bá***(ra)***soma*

(9d) NEG

ntiba**(ra)***somá*

extra segments hinder analogy

Periphrastic negation uses relativization

(10a) *Ntaabwo ba(ra)somá.*

NEG they.are.about.to.read.**REL**

'They are not about to read.'

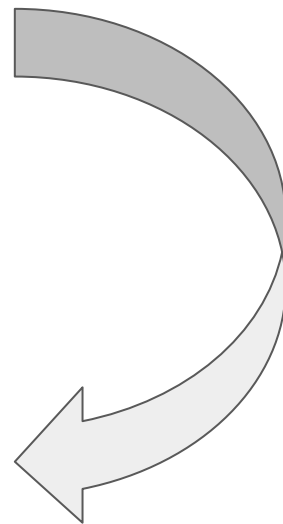
(early change?)

(10b) *Nti-ba(ra)somá.*

NEG-they.are.about.to.read

'They are not about to read.'

(late change?)



additional
pressure
created by
REL?

surface similarity
facilitates analogical
morphological
change

→ currently testing this!
how? ask me / see handout!

why did one "settle" higher than the other?

61%



41%



frequent infrequent



frequency facilitates
leveling of competing
morphosyntactic
strategies (Bybee 1995,
Smith 2001)

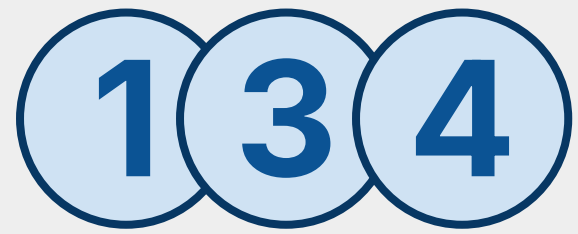
maybe:

frequency facilitates
greater use if variation
reaches stability?

wrapping up

Rwandans use *ra-*
differently!

wrapping up



some *ra*- variation is
stable



wrapping up

ra- is **expanding into**
new syntactic
environments



wrapping up

surface similarity
facilitates analogical
change?



wrapping up

frequency facilitates
higher use in stable
variation?

Murakoze!

'Thank you!'

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